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PROJECT SYNOPSIS

AI-ENABLED AUTOSAR HLD DOCUMENT ANALYSIS ASSISTANT

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ABSTRACT

AUTOSAR High-Level Design (HLD) documents have valuable information related to the components, interfaces, ports, signals, dependencies, and functional interactions in automotive software systems. Manually analyzing these documents can take up time, especially when architectural information is scattered across long technical documents. Finding the information and making sense of the relationship between the architecture elements could further increase the time needed for review and integration of the design. For addressing these issues, we present the AI-Powered AUTOSAR HLD Document Analysis Assistant that allows users to upload PDF documents and query any information present in these documents using natural language.The solution uses the Retrieval-Augmented Generation (RAG) framework for generating document-grounded responses. The frontend of the application is developed using React and Vite, while the backend is built using Python and Flask for handling document uploads, processing requests, and answering questions on the documents. PyMuPDF library is used for extracting textual information from the uploaded PDF documents. Text extracted from the documents is split into small chunks and represented as vector embeddings using the MiniLM embedding model. The vector embeddings are indexed using Facebook AI Similarity Search (FAISS).The documents are used as background information to the Qwen2.5 3B language model to generate natural language responses for query questions. The meta-information about the documents is stored in a SQLite database while the uploaded PDF documents, extracted texts and index information are stored in the filesystem.The developed assistant is designed to reduce the effort spent on documents search, access the architectural documents, and assist the engineers in design analysis and information retrieval.

Signature: D. Bhuvaneshwari

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